

Demographic Analysis: Japan & U.S. Census

Shuja Waraich

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1 Analysis of Japanese Demographic Data

1.1 Data Loading and Preparation

```
# Load subdivisions data (population by year)
subdivisions <- read_excel("japan-subdivisions.xlsx") %>%
  select(Name, Status, pop1995:pop2020)

# Load prefecture areas data (skip the blank first row)
areas_raw <- read_excel("japan-prefecture-areas.xlsx", skip = 2)

# Clean column names for the areas file
names(areas_raw) <- c("Rank", "Prefecture", "Japanese", "EstArea_km2",
  "DetArea_km2", "UndetArea_km2", "Population",
  "Density_per_km2", "InhabArea_km2",
  "InhabDensity_per_km2", "ForestPct", "Capital")

areas <- areas_raw %>%
  filter(!is.na(Rank)) %>%
  mutate(across(c(Rank, EstArea_km2, DetArea_km2, Population,
    Density_per_km2), as.numeric))
```

1.2 Prefecture-Level Population Percentage Changes (1995–2020)

```
# Filter for prefectures only
prefectures <- subdivisions %>%
  filter(Status == "Prefecture") %>%
  mutate(across(pop1995:pop2020, as.numeric))

# Calculate percentage change relative to 1995
pref_change <- prefectures %>%
  mutate(
    pct_2000 = (pop2000 - pop1995) / pop1995 * 100,
    pct_2005 = (pop2005 - pop1995) / pop1995 * 100,
    pct_2010 = (pop2010 - pop1995) / pop1995 * 100,
    pct_2015 = (pop2015 - pop1995) / pop1995 * 100,
    pct_2020 = (pop2020 - pop1995) / pop1995 * 100
  )

# Display summary table
pref_change %>%
  select(Name, pop1995, pop2020, pct_2020) %>%
  arrange(desc(pct_2020)) %>%
  mutate(pop1995 = comma(pop1995),
    pop2020 = comma(pop2020),
    pct_2020 = round(pct_2020, 2)) %>%
  rename(`Prefecture` = Name,
    `Pop. 1995` = pop1995,
    `Pop. 2020` = pop2020,
    `% Change (1995-2020)` = pct_2020) %>%
```

```

kable(caption = "Population Change by Prefecture (1995--2020)",
      booktabs = TRUE) %>%
kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
              font_size = 8)

```

Table 1: Population Change by Prefecture (1995–2020)

Prefecture	Pop. 1995	Pop. 2020	% Change (1995-2020)
Okinawa	1,273,440	1,467,480	15.24
Kanagawa	8,245,903	9,237,337	12.02
Shiga	1,287,005	1,413,610	9.84
Aichi	6,868,336	7,542,415	9.81
Saitama	6,759,311	7,344,765	8.66
Chiba	5,797,782	6,284,480	8.39
Fukuoka	4,933,393	5,135,214	4.09
Hyōgo	5,401,877	5,465,002	1.17
Osaka	8,797,268	8,837,685	0.46
Miyagi	2,328,739	2,301,996	-1.15
Kyōto	2,629,592	2,578,087	-1.96
Tochigi	1,984,390	1,933,146	-2.58
Shizuoka	3,737,689	3,633,202	-2.80
Hiroshima	2,881,748	2,799,702	-2.85
Ibaraki	2,955,530	2,867,009	-3.00
Okayama	1,950,750	1,888,432	-3.19
Gumma [Gunma]	2,003,540	1,939,110	-3.22
Mie	1,841,358	1,770,254	-3.86
Ishikawa	1,180,068	1,132,526	-4.03
Gifu	2,102,442	1,978,742	-5.88
Kumamoto	1,859,793	1,738,301	-6.53
Nagano	2,191,857	2,048,011	-6.56
Fukui	826,996	766,863	-7.27
Nara	1,430,862	1,324,473	-7.44
Kagawa	1,027,006	950,244	-7.47
Toyama	1,123,125	1,034,814	-7.86
Yamanashi	881,996	809,974	-8.17
Hokkaidō	5,692,321	5,224,614	-8.22
Saga	884,316	811,442	-8.24
Ōita	1,231,306	1,123,852	-8.73
Miyazaki	1,175,819	1,069,576	-9.04
Tottori	614,929	553,407	-10.00
Ehime	1,506,700	1,334,841	-11.41
Kagoshima	1,794,224	1,588,256	-11.48
Niigata	2,488,364	2,201,272	-11.54
Shimane	771,441	671,126	-13.00
Tokushima	832,427	719,559	-13.56
Yamaguchi	1,555,543	1,342,059	-13.72
Fukushima	2,133,592	1,833,152	-14.08
Wakayama	1,080,435	922,584	-14.61
Iwate	1,419,505	1,210,534	-14.72
Yamagata	1,256,958	1,068,027	-15.03
Nagasaki	1,544,934	1,312,317	-15.06
Kōchi	816,704	691,527	-15.33
Aomori	1,481,663	1,237,984	-16.45
Akita	1,213,667	959,502	-20.94

1.3 Time Series Plot: Percentage Change Relative to 1995

```
# Reshape to long format for plotting
pref_long <- pref_change %>%
  select(Name, starts_with("pct_")) %>%
  pivot_longer(cols = starts_with("pct_"),
               names_to = "year",
               values_to = "pct_change") %>%
  mutate(year = as.numeric(str_extract(year, "\\d{4}")))

# Add the baseline year (1995 = 0%)
baseline <- prefectures %>%
  select(Name) %>%
  mutate(year = 1995, pct_change = 0)

pref_long <- bind_rows(baseline, pref_long)

# Identify top gainers and losers for labeling
extremes <- pref_change %>%
  arrange(desc(pct_2020)) %>%
  slice(c(1:3, (n()-2):n())) %>%
  pull(Name)

ggplot(pref_long, aes(x = year, y = pct_change, group = Name)) +
  geom_line(data = filter(pref_long, !Name %in% extremes),
            alpha = 0.3, color = "grey60", linewidth = 0.4) +
  geom_line(data = filter(pref_long, Name %in% extremes),
            aes(color = Name), linewidth = 0.9) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  scale_x_continuous(breaks = seq(1995, 2020, 5)) +
  labs(x = "Year", y = "% Change Relative to 1995",
       title = "Population Change by Prefecture (1995--2020)",
       color = "Prefecture") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

The plot highlights the top 3 growing and bottom 3 declining prefectures in color, with all other prefectures shown in grey. Most prefectures experienced population decline since 1995, while a handful of urban prefectures (e.g., Tokyo-area) grew.

1.4 Scatter Plot: 1995 Population vs. 1995–2020 Percentage Change

```
ggplot(pref_change, aes(x = pop1995, y = pct_2020)) +
  geom_point(color = "steelblue", size = 2.5) +
  geom_smooth(method = "lm", se = TRUE, color = "firebrick", linetype = "dashed") +
  geom_hline(yintercept = 0, linetype = "dotted") +
  scale_x_continuous(labels = comma) +
  labs(x = "Population in 1995",
       y = "% Change (1995--2020)",
       title = "Larger Prefectures Tend to Grow (or Decline Less)") +
  theme_minimal()
```

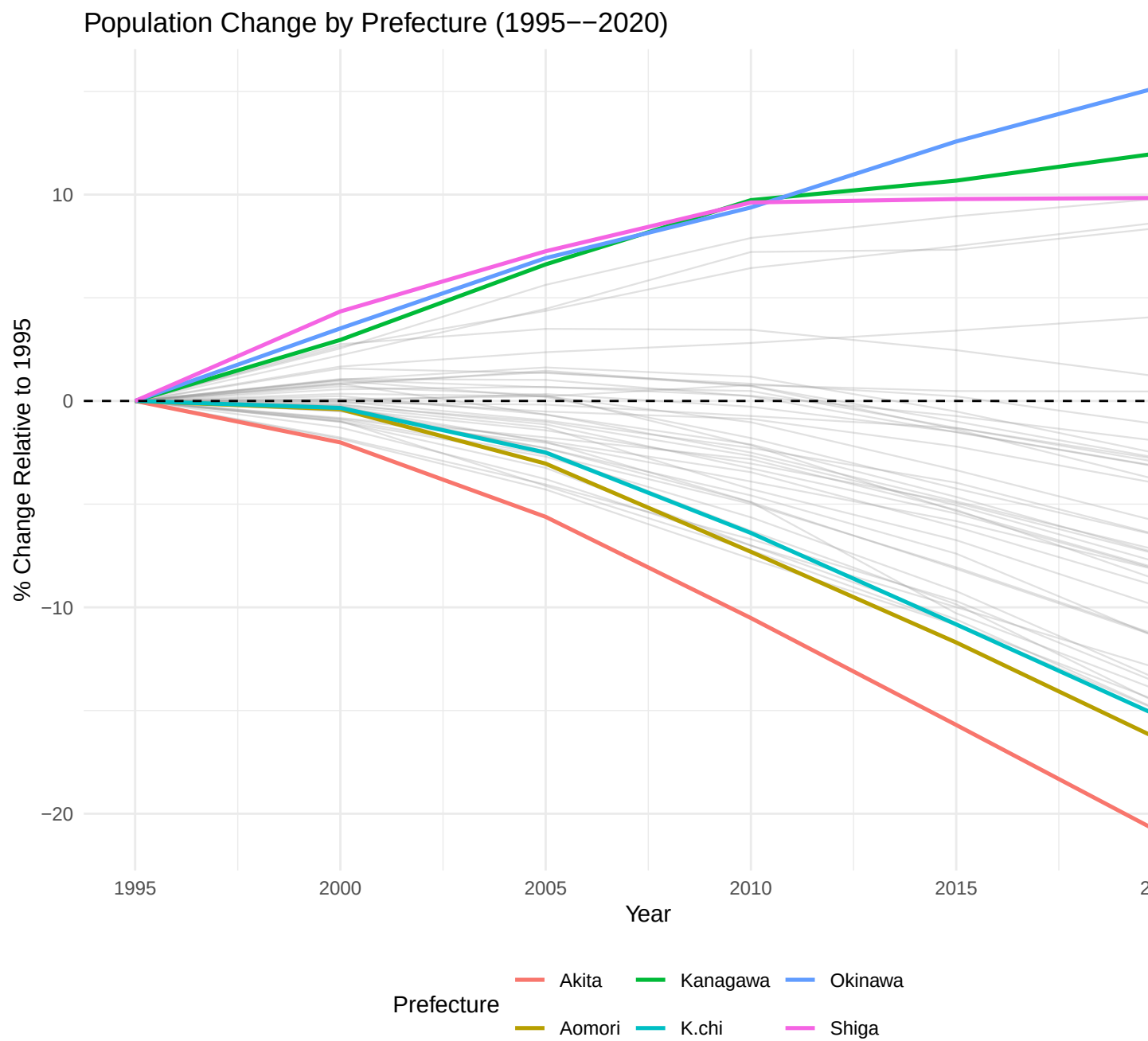


Figure 1: Percentage Population Change Relative to 1995 by Prefecture

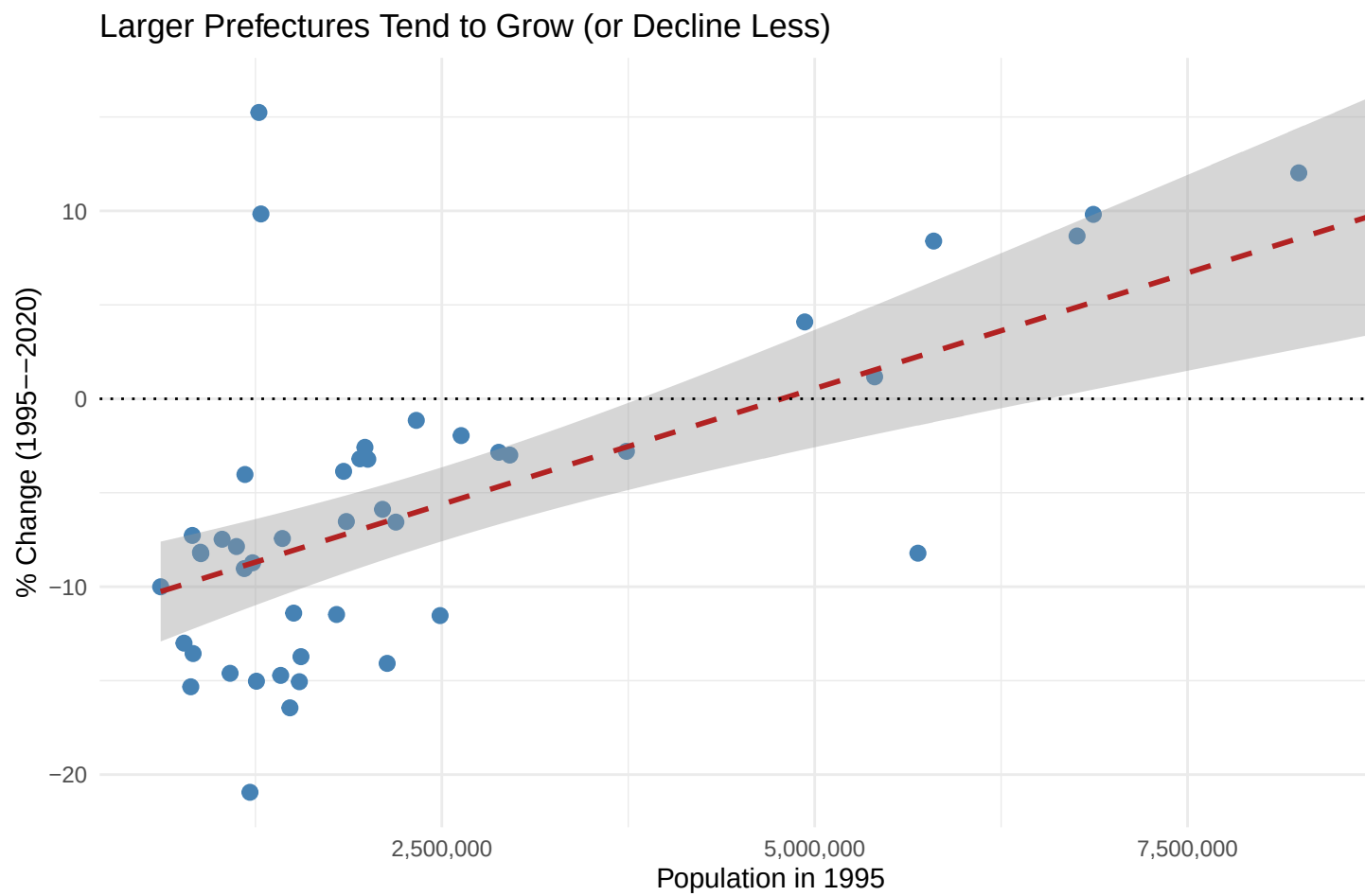


Figure 2: 1995 Population vs. Percentage Change (1995–2020)

```
cor_val <- cor(pref_change$pop1995, pref_change$pct_2020)
```

Correlation coefficient (Pearson): 0.615. There is a positive relationship between 1995 population size and subsequent growth—larger prefectures tended to grow or at least decline less than smaller ones.

1.5 City-Level Analysis: Does City Size Relate to Population Change?

```
# Filter for cities only
cities <- subdivisions %>%
  filter(str_detect(Status, "City")) %>%
  mutate(across(pop1995:pop2020, as.numeric)) %>%
  filter(!is.na(pop1995), pop1995 > 0) %>%
  mutate(pct_change = (pop2020 - pop1995) / pop1995 * 100)

# Categorize city size
cities <- cities %>%
  mutate(size_cat = case_when(
    pop1995 >= 1000000 ~ "Large (1M+)",
    pop1995 >= 500000 ~ "Medium (500K-1M)",
    pop1995 >= 100000 ~ "Small (100K-500K)",
    TRUE ~ "Town (<100K)"
  ),
  size_cat = factor(size_cat, levels = c("Town (<100K)", "Small (100K-500K)",
    "Medium (500K-1M)", "Large (1M+)")))

ggplot(cities, aes(x = pop1995, y = pct_change)) +
  geom_point(aes(color = size_cat), alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, color = "black", linetype = "dashed") +
  geom_hline(yintercept = 0, linetype = "dotted") +
  scale_x_log10(labels = comma) +
  labs(x = "City Population in 1995 (log scale)",
    y = "% Change (1995--2020)",
    title = "City Size and Population Change",
    color = "City Size") +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```
cities %>%
  group_by(size_cat) %>%
  summarise(
    n = n(),
    mean_change = round(mean(pct_change, na.rm = TRUE), 2),
    median_change = round(median(pct_change, na.rm = TRUE), 2),
    .groups = "drop"
  ) %>%
  rename(`City Size Category` = size_cat,
    `Count` = n,
    `Mean % Change` = mean_change,
    `Median % Change` = median_change) %>%
  kable(caption = "Population Change by City Size Category",
```

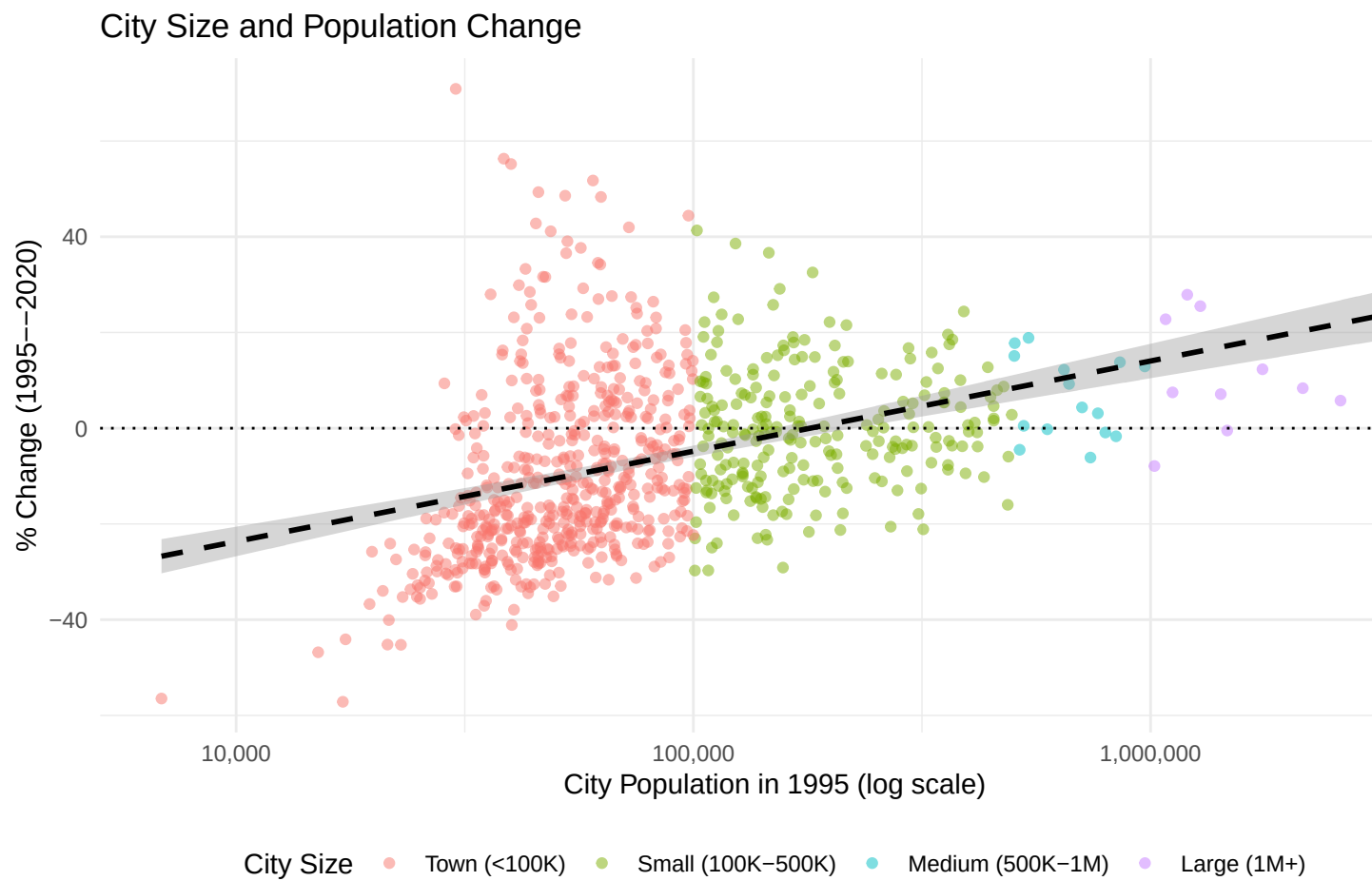


Figure 3: City Population in 1995 vs. Percentage Change (1995-2020)


```
booktabs = TRUE) %>%
kable_styling(latex_options = c("striped", "hold_position"))
```

Table 2: Population Change by City Size Category

City Size Category	Count	Mean % Change	Median % Change
Town (<100K)	530	-9.93	-14.48
Small (100K-500K)	236	-0.25	-0.83
Medium (500K-1M)	15	6.29	4.35
Large (1M+)	11	11.18	8.36

Larger cities tend to maintain or grow their population, while smaller towns and cities are more likely to experience decline.

1.6 Correlation: Population Density vs. Percentage Growth

```
# Merge prefecture changes with area/density data
density_df <- pref_change %>%
  left_join(areas, by = c("Name" = "Prefecture")) %>%
  filter(!is.na(Density_per_km2))

ggplot(density_df, aes(x = Density_per_km2, y = pct_2020)) +
  geom_point(color = "darkorange", size = 2.5) +
  geom_smooth(method = "lm", se = TRUE, color = "firebrick", linetype = "dashed") +
  geom_hline(yintercept = 0, linetype = "dotted") +
  scale_x_log10() +
  labs(x = "Population Density (per km², log scale)",
       y = "% Change (1995--2020)",
       title = "Denser Prefectures Show Greater Population Resilience") +
  theme_minimal()
```

```
density_cor <- cor(log(density_df$Density_per_km2), density_df$pct_2020,
                   use = "complete.obs")
```

Correlation (log-density vs. % change): 0.837. Higher population density is positively associated with population growth, confirming that urbanized prefectures are better at retaining and attracting residents.

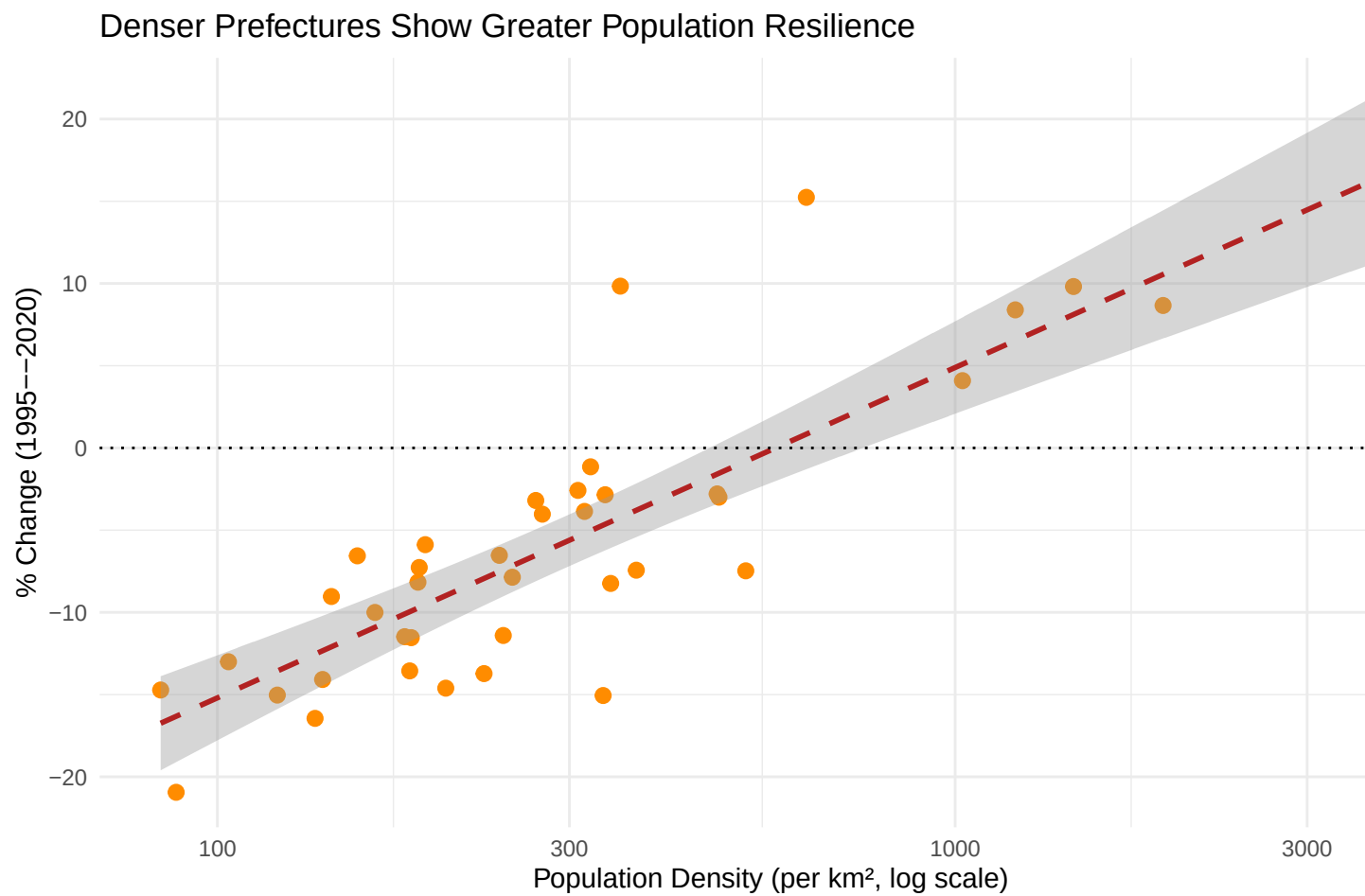


Figure 4: Population Density vs. Percentage Growth (1995–2020)

2 U.S. Census & TidyCensus Analysis

```
library(tidycensus)
library(sf)
library(tigris)
library(viridis)
options(tigris_use_cache = TRUE)
census_api_key("cae79ee00ab828454493cdb63e0d615494c2a5e", install = FALSE)
```

2.1 Population Change: County-Level 2000 to 2020

```
# 2000 Decennial Census: P001001 = Total Population (SF1)
pop_2000 <- get_decennial(
  geography = "county",
  variables = "P001001",
  year = 2000,
  sumfile = "sf1"
) %>%
  rename(pop2000 = value) %>%
  select(GEOID, NAME, pop2000)

# 2020 Decennial Census: P1_001N = Total Population (PL 94-171)
pop_2020 <- get_decennial(
  geography = "county",
  variables = "P1_001N",
  year = 2020
) %>%
  rename(pop2020 = value) %>%
  select(GEOID, NAME, pop2020)

# Merge and calculate percentage change
us_county_change <- pop_2000 %>%
  inner_join(pop_2020, by = "GEOID") %>%
  mutate(pct_change = (pop2020 - pop2000) / pop2000 * 100) %>%
  rename(NAME = NAME.x) %>%
  select(GEOID, NAME, pop2000, pop2020, pct_change)

us_county_change %>%
  arrange(desc(pct_change)) %>%
  slice(c(1:5, (n()-4):n())) %>%
  mutate(across(c(pop2000, pop2020), comma),
    pct_change = round(pct_change, 1)) %>%
  kable(caption = "Top 5 and Bottom 5 Counties by Population Change (2000--2020)",
    booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"))
```

Table 3: Top 5 and Bottom 5 Counties by Population Change (2000–2020)

GEOID	NAME	pop2000	pop2020	pct_change
46083	Lincoln County, South Dakota	24,131	65,161	170.0
38053	McKenzie County, North Dakota	5,737	14,704	156.3
13117	Forsyth County, Georgia	98,407	251,283	155.4
48397	Rockwall County, Texas	43,080	107,819	150.3
51107	Loudoun County, Virginia	169,599	420,959	148.2
28055	Issaquena County, Mississippi	2,274	1,338	-41.2
28125	Sharkey County, Mississippi	6,580	3,800	-42.2
22023	Cameron Parish, Louisiana	9,991	5,617	-43.8
15005	Kalawao County, Hawaii	147	82	-44.2
17003	Alexander County, Illinois	9,590	5,240	-45.4

2.2 Variable Verification

P001001 (2000 Decennial Census): This variable comes from **Summary File 1 (SF1)**, Table P1 (“Total Population”). It provides the complete count of all persons enumerated in the 2000 Census.

P1_001N (2020 Decennial Census): This variable comes from the **PL 94-171 Redistricting Data** file, Table P1 (“Race”). The _001N suffix indicates the total population count. This is the primary redistricting dataset released from the 2020 Census.

2.3 U.S. Scatter Plots

2.3.1 Growth Rate vs. Total Population

```
ggplot(us_county_change, aes(x = pop2000, y = pct_change)) +
  geom_point(alpha = 0.3, size = 0.8, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "firebrick") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  scale_x_log10(labels = comma) +
  coord_cartesian(ylim = c(-75, 200)) +
  labs(x = "Total Population in 2000 (log scale)",
       y = "% Change (2000--2020)",
       title = "County Population Growth vs. Size") +
  theme_minimal()
```

2.3.2 RUCC Classification

```
# Load USDA Rural-Urban Continuum Codes (pre-downloaded XLS)
rucc <- read_excel("rucc2013.xls") %>%
  mutate(FIPS = str_pad(FIPS, 5, pad = "0")) %>%
  select(FIPS, RUCC_2013, Description)

# Merge with county change data
us_rucc <- us_county_change %>%
  left_join(rucc, by = c("GEOID" = "FIPS")) %>%
  filter(!is.na(RUCC_2013)) %>%
  mutate(
```



Figure 5: Growth Rate vs. Population for U.S. Counties

```

RUCC_2013 = factor(RUCC_2013),
metro_status = ifelse(as.numeric(as.character(RUCC_2013)) <= 3,
                      "Metro", "Nonmetro")
)

```

```

ggplot(us_rucc, aes(x = pop2000, y = pct_change, color = RUCC_2013)) +
  geom_point(alpha = 0.4, size = 0.9) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  scale_x_log10(labels = comma) +
  scale_color_viridis_d(option = "turbo") +
  coord_cartesian(ylim = c(-75, 200)) +
  labs(x = "Population in 2000 (log scale)",
       y = "% Change (2000--2020)",
       title = "County Growth by RUCC Code",
       color = "RUCC") +
  theme_minimal() +
  theme(legend.position = "right")

```

2.3.3 Metro vs. Nonmetro Classification

```

ggplot(us_rucc, aes(x = pop2000, y = pct_change, color = metro_status)) +
  geom_point(alpha = 0.3, size = 0.9) +
  geom_smooth(method = "lm", se = TRUE) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  scale_x_log10(labels = comma) +
  scale_color_manual(values = c("Metro" = "#2166AC", "Nonmetro" = "#B2182B")) +
  coord_cartesian(ylim = c(-75, 200)) +
  labs(x = "Population in 2000 (log scale)",
       y = "% Change (2000--2020)",
       title = "Metro vs. Nonmetro County Growth",
       color = "Classification") +
  theme_minimal()

```

Metro counties tend to grow faster on average, while nonmetro counties are more likely to experience population decline.

2.4 Population Pyramids

2.4.1 Metro vs. Nonmetro in Illinois

```

# Get age/sex data for Illinois counties from ACS 2020
il_age_vars <- paste0("B01001_", str_pad(2:49, 3, pad = "0"))

il_agesex <- get_acs(
  geography = "county",
  variables = il_age_vars,
  state = "IL",
  year = 2020,

```

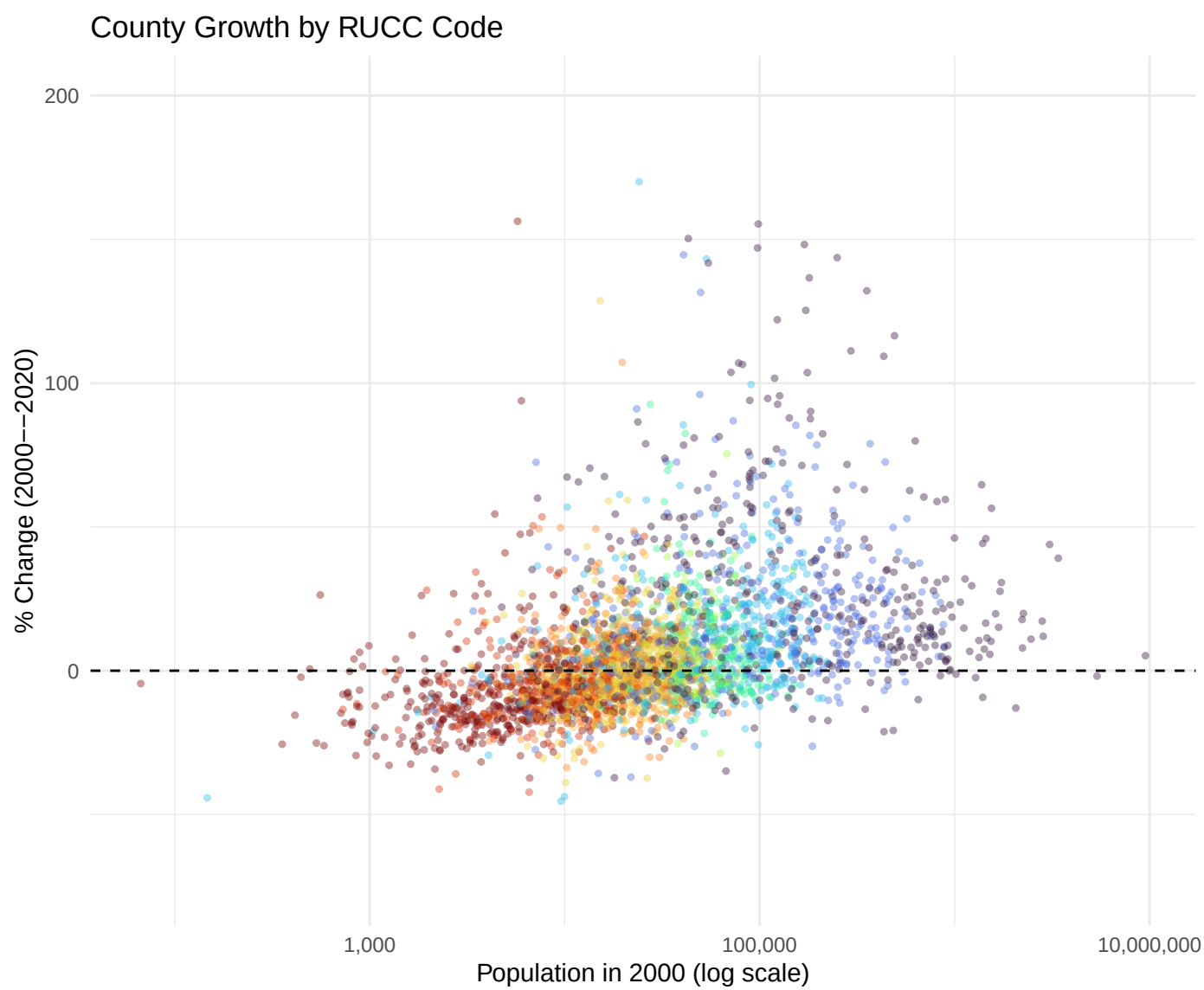


Figure 6: Growth Rate Colored by RUCC Code

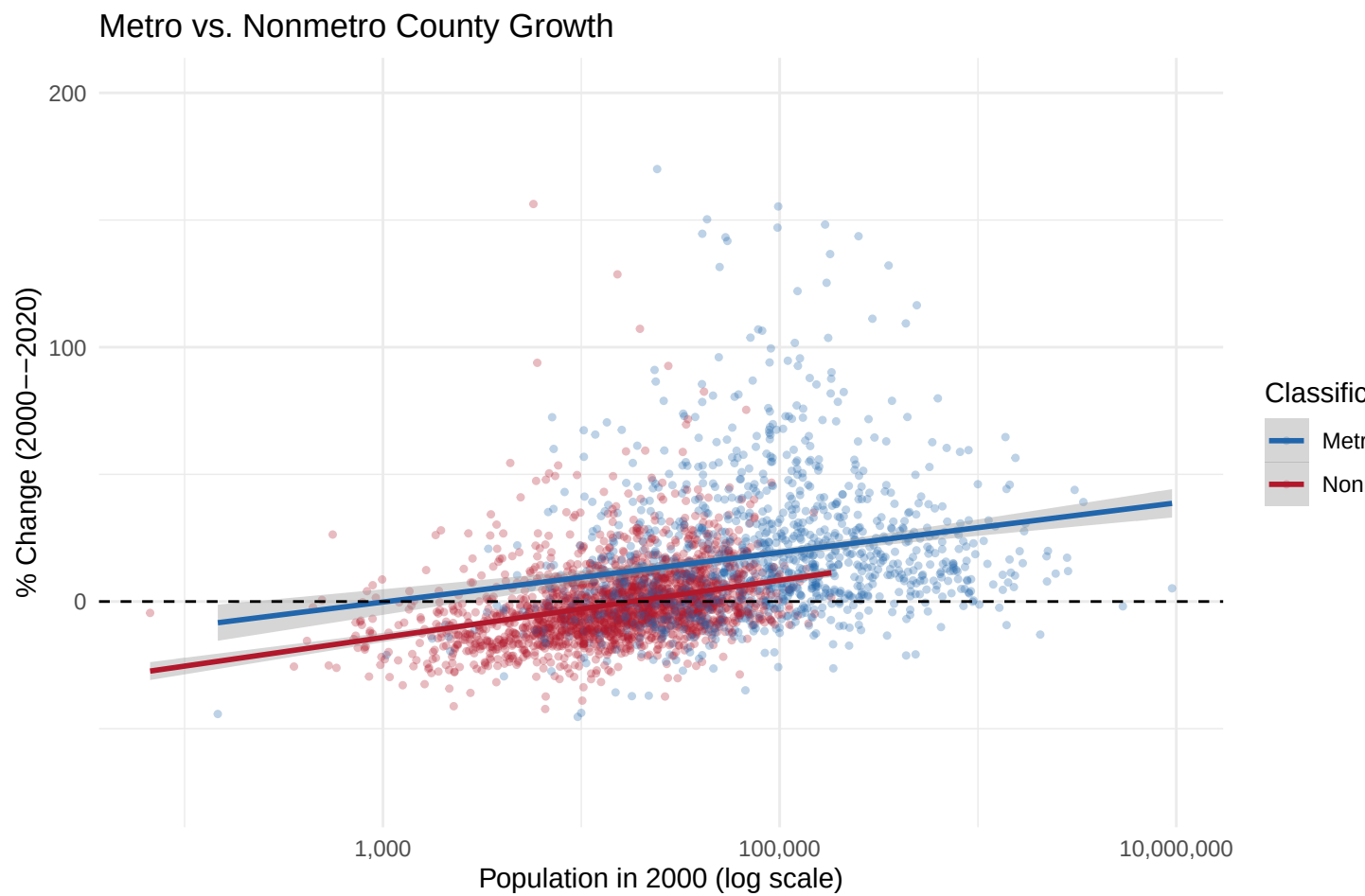


Figure 7: Growth Rate: Metro (RUCC 1-3) vs. Nonmetro (RUCC 4-9)


```

  survey = "acs5"
)

# Map variable codes to age groups and sex
# B01001_002 = Male total, B01001_003-025 = Male age groups
# B01001_026 = Female total, B01001_027-049 = Female age groups
age_labels <- c(
  "Under 5", "5-9", "10-14", "15-17", "18-19", "20", "21",
  "22-24", "25-29", "30-34", "35-39", "40-44", "45-49",
  "50-54", "55-59", "60-61", "62-64", "65-66", "67-69",
  "70-74", "75-79", "80-84", "85+"
)

male_vars <- paste0("B01001_", str_pad(3:25, 3, pad = "0"))
female_vars <- paste0("B01001_", str_pad(27:49, 3, pad = "0"))

male_map <- tibble(variable = male_vars, age = age_labels, sex = "Male")
female_map <- tibble(variable = female_vars, age = age_labels, sex = "Female")
var_map <- bind_rows(male_map, female_map)

il_pyramid <- il_agesex %>%
  inner_join(var_map, by = "variable") %>%
  left_join(rucc %>% rename(GEOID = FIPS), by = "GEOID") %>%
  filter(!is.na(RUCC_2013)) %>%
  mutate(metro_status = ifelse(RUCC_2013 <= 3, "Metro", "Nonmetro"))

# Aggregate age groups into broader bins for readability
il_pyramid <- il_pyramid %>%
  mutate(age_group = case_when(
    age %in% c("Under 5", "5-9") ~ "0-9",
    age %in% c("10-14", "15-17") ~ "10-17",
    age %in% c("18-19", "20", "21", "22-24") ~ "18-24",
    age %in% c("25-29", "30-34") ~ "25-34",
    age %in% c("35-39", "40-44") ~ "35-44",
    age %in% c("45-49", "50-54") ~ "45-54",
    age %in% c("55-59", "60-61", "62-64") ~ "55-64",
    age %in% c("65-66", "67-69", "70-74") ~ "65-74",
    age %in% c("75-79", "80-84", "85+") ~ "75+"
  )) %>%
  mutate(age_group = factor(age_group,
    levels = c("0-9", "10-17", "18-24", "25-34", "35-44",
      "45-54", "55-64", "65-74", "75+")))

il_pyr_agg <- il_pyramid %>%
  group_by(metro_status, sex, age_group) %>%
  summarise(pop = sum(estimate, na.rm = TRUE), .groups = "drop") %>%
  group_by(metro_status) %>%
  mutate(pct = pop / sum(pop) * 100) %>%
  ungroup() %>%
  mutate(pct = ifelse(sex == "Male", -pct, pct))

ggplot(il_pyr_agg, aes(x = age_group, y = pct, fill = sex)) +
  geom_bar(stat = "identity") +

```

```
coord_flip() +
facet_wrap(~ metro_status) +
scale_fill_manual(values = c("Male" = "#4393C3", "Female" = "#D6604D")) +
scale_y_continuous(labels = function(x) paste0(abs(round(x, 1)), "%")) +
labs(x = "Age Group", y = "% of Population",
     title = "Illinois Population Pyramids: Metro vs. Nonmetro",
     fill = "Sex") +
theme_minimal() +
theme(legend.position = "bottom")
```

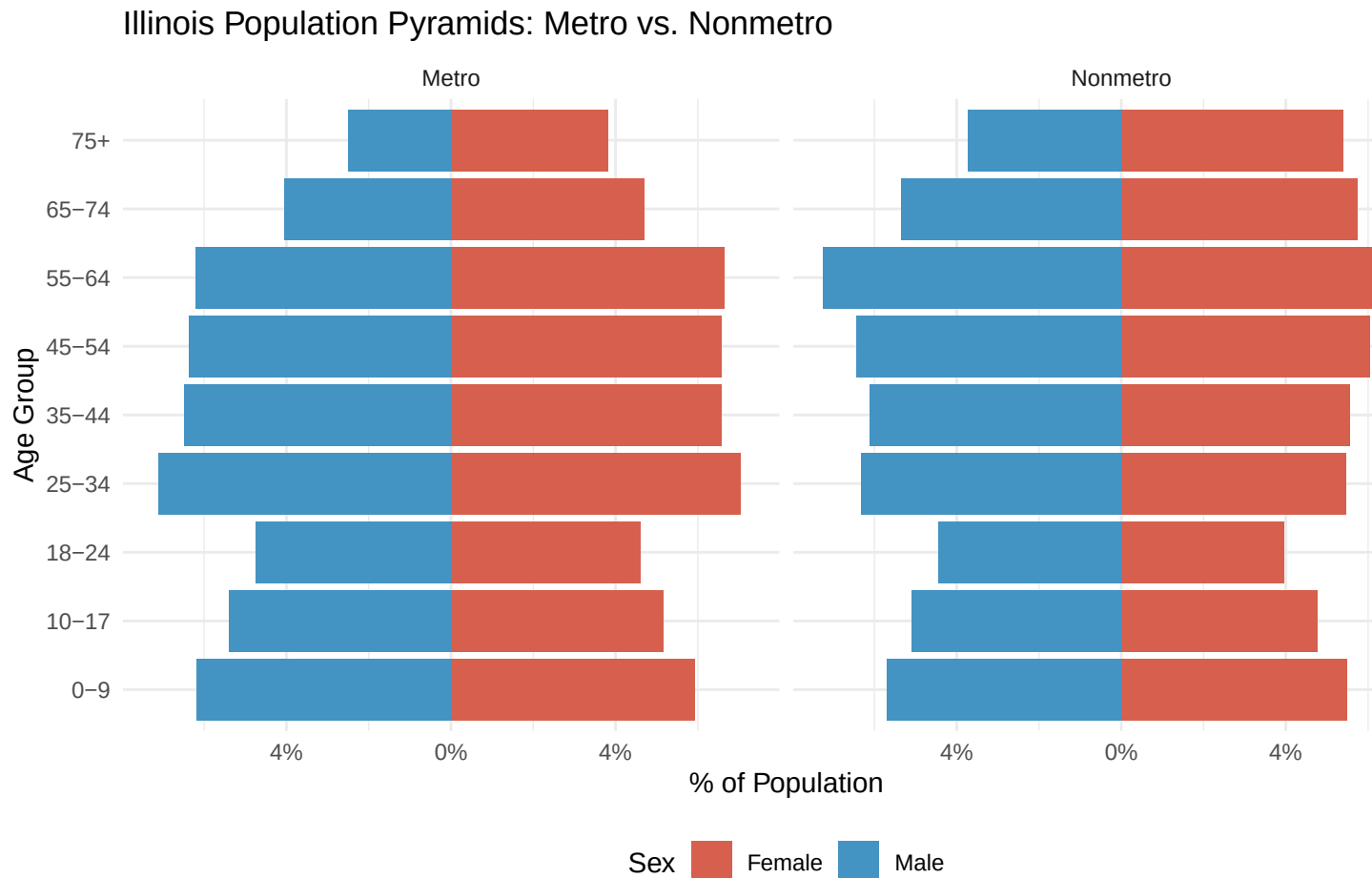


Figure 8: Population Pyramids: Metro vs. Nonmetro Counties in Illinois

2.4.2 Three-Way Comparison: Metro, Micro, and Rural in the Midwest

```
midwest_states <- c("IL", "IN", "IA", "KS", "MI", "MN", "MO",
                    "NE", "ND", "OH", "SD", "WI")

mw_agesex <- get_acs(
  geography = "county",
```

```

variables = il_age_vars,
state = midwest_states,
year = 2020,
survey = "acs5"
)

mw_pyramid <- mw_agesex %>%
  inner_join(var_map, by = "variable") %>%
  left_join(rucc %>% rename(GEOID = FIPS), by = "GEOID") %>%
  filter(!is.na(RUCC_2013)) %>%
  mutate(
    urban_class = case_when(
      RUCC_2013 <= 3 ~ "Metro",
      RUCC_2013 <= 7 ~ "Micro",
      TRUE ~ "Rural"
    ),
    urban_class = factor(urban_class, levels = c("Metro", "Micro", "Rural"))
  ) %>%
  mutate(age_group = case_when(
    age %in% c("Under 5", "5-9") ~ "0-9",
    age %in% c("10-14", "15-17") ~ "10-17",
    age %in% c("18-19", "20", "21", "22-24") ~ "18-24",
    age %in% c("25-29", "30-34") ~ "25-34",
    age %in% c("35-39", "40-44") ~ "35-44",
    age %in% c("45-49", "50-54") ~ "45-54",
    age %in% c("55-59", "60-61", "62-64") ~ "55-64",
    age %in% c("65-66", "67-69", "70-74") ~ "65-74",
    age %in% c("75-79", "80-84", "85+") ~ "75+"
  )) %>%
  mutate(age_group = factor(age_group,
    levels = c("0-9", "10-17", "18-24", "25-34", "35-44",
      "45-54", "55-64", "65-74", "75+")))

mw_pyr_agg <- mw_pyramid %>%
  group_by(urban_class, sex, age_group) %>%
  summarise(pop = sum(estimate, na.rm = TRUE), .groups = "drop") %>%
  group_by(urban_class) %>%
  mutate(pct = pop / sum(pop) * 100) %>%
  ungroup() %>%
  mutate(pct = ifelse(sex == "Male", -pct, pct))

ggplot(mw_pyr_agg, aes(x = age_group, y = pct, fill = sex)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_wrap(~ urban_class, ncol = 3) +
  scale_fill_manual(values = c("Male" = "#4393C3", "Female" = "#D6604D")) +
  scale_y_continuous(labels = function(x) paste0(abs(round(x, 1)), "%")) +
  labs(x = "Age Group", y = "% of Population",
    title = "Midwest Population Pyramids by Urbanization Level",
    fill = "Sex") +
  theme_minimal() +
  theme(legend.position = "bottom")

```

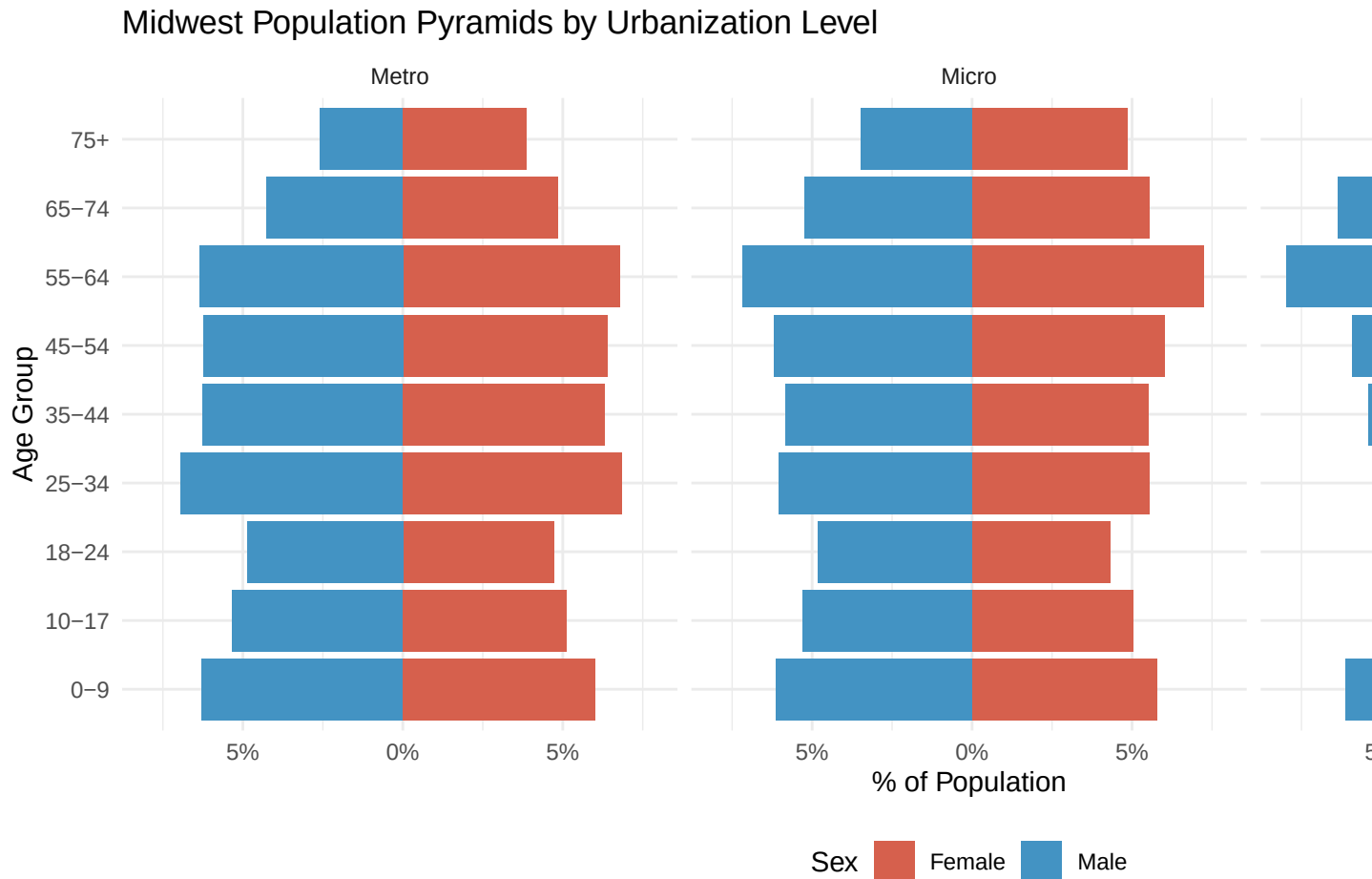


Figure 9: Midwest Population Pyramids: Metro vs. Micro vs. Rural

Rural areas show a notably older age structure with proportionally fewer working-age adults (25–44), consistent with youth outmigration to metro areas.

2.5 Geospatial Mapping: Midwest County Population Change

```
# Get county geometries for Midwest states
mw_fips <- c("17","18","19","20","26","27","29","31","38","39","46","55")

mw_counties <- counties(state = mw_fips, year = 2020, cb = TRUE) %>%
  st_transform(crs = 5070)

## |

# Merge with population change data
mw_map_data <- mw_counties %>%
  left_join(us_county_change, by = "GEOID") %>%
  mutate(pct_change_capped = pmin(pmax(pct_change, -50), 50))

ggplot(mw_map_data) +
  geom_sf(aes(fill = pct_change_capped), color = "grey80", linewidth = 0.05) +
  scale_fill_gradient2(
    low = "#B2182B", mid = "white", high = "#2166AC",
    midpoint = 0,
    limits = c(-50, 50),
    name = "% Change\n(2000-2020)",
    labels = function(x) ifelse(abs(x) == 50, paste0(x, "+"), x)
  ) +
  labs(title = "Population Change in Midwest Counties (2000--2020)") +
  theme_void() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
    legend.position = "right"
  )
```

The choropleth reveals a stark urban-rural divide: metropolitan cores and suburban rings (blue) grew, while vast swaths of rural counties (red) experienced significant population loss over the past two decades.

Population Change in Midwest Counties (2000--2020)

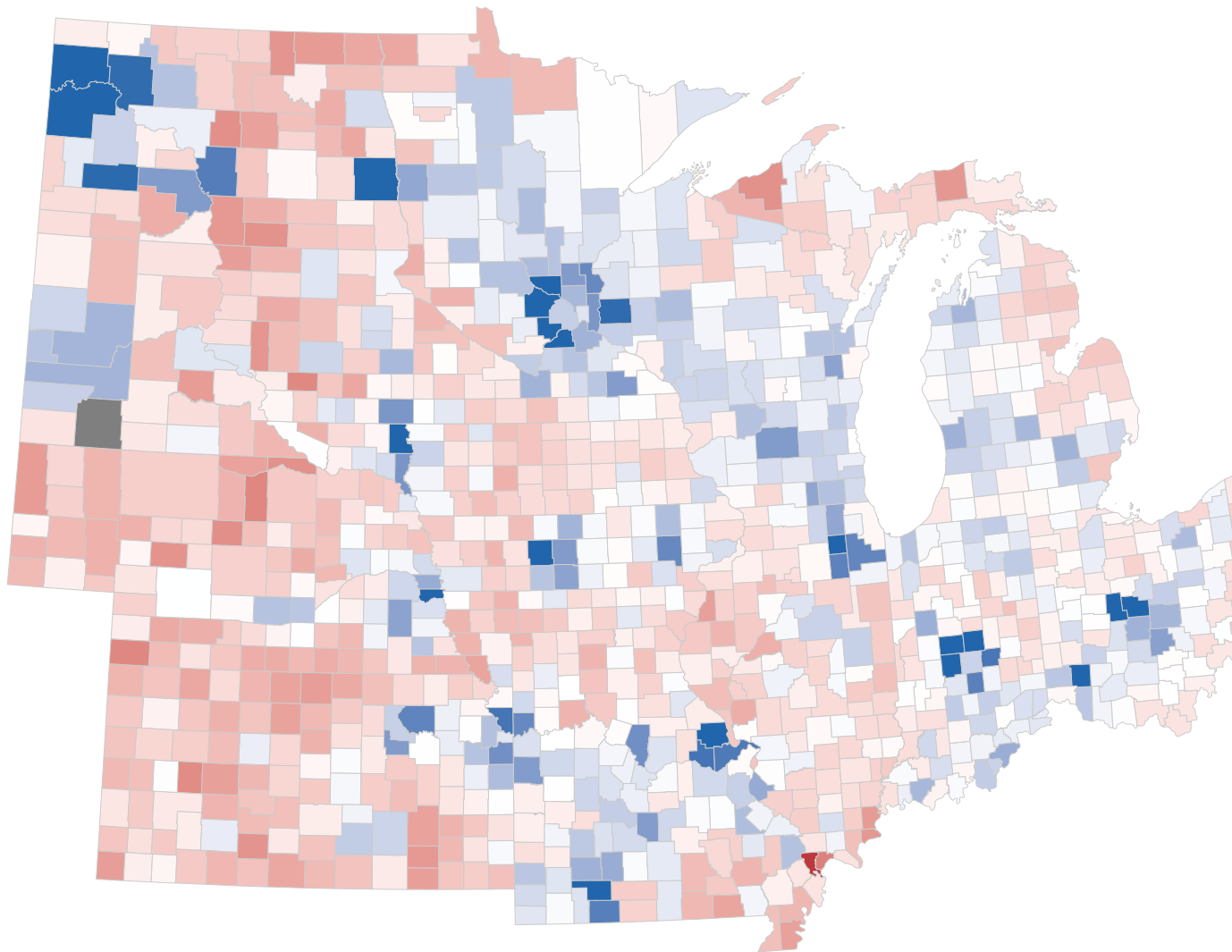


Figure 10: Percentage Population Change (2000–2020) in Midwest Counties